

# REGINA H REYNOLDS

Molecular biologist turned bioinformatician, with experience translating large-scale omics data into actionable insights for drug discovery. I've led cross-functional analyses across transcriptomics, genomics, and functional screens, collaborating closely across teams to build reproducible pipelines that support impactful R&D decisions. I'm driven by a commitment to scientific rigour, integrity, and transparency – and by a broader desire to make a meaningful contribution to the world through thoughtful, data-driven science.

View this CV online with links at <https://rhreynolds.github.io/cv>



## WORK EXPERIENCE

Present  
|  
2025

### Senior Principal Scientist, Bioinformatics Lead

AstronauTx

📍 London, UK

- Leading development of a modular EEG analytics platform to bring EEG analysis in-house in support of AstronauTx's sleep-based therapeutic approach. Built in Python and Nextflow, with each processing stage independently tested and validated against reference outputs.
- Contributing to translational understanding and biomarker strategy through QC, harmonisation, and analysis of historical and incoming EEG and other assay data.
- Establishing AstronauTx's computational foundations, spanning: cloud infrastructure; engineering practices; company-wide onboarding to Claude with guidance on output evaluation; and automation of manual processes (e.g. daily synchronisation of experimental timelines from the electronic lab notebook system into Monday.com via GitHub Actions).

2025  
|  
2022

### Lead Computational Biologist

CoSyne Therapeutics

📍 London, UK

- Provided scientific and strategic leadership as lead of a 5-person computational biology team, translating high-level objectives into clear milestones and actionable plans, managing tight deadlines and resources, and setting research priorities in collaboration with team members, experimental, engineering, and AI teams, and senior leadership.
- Drove company-wide 'omics initiatives spanning whole-genome sequencing, transcriptomics and CRISPRi screens to support target discovery in glioblastoma multiforme.
- Established scalable, robust, and reproducible computational workflows through adoption of tools such as Nextflow and Seqera Cloud for workflow orchestration, Docker for containerisation, R targets for reproducible exploratory data analysis, and Cruft templates for project scaffolding.
- Contributed to shaping company culture through line management; growing the computational biology team from 2 to 5 people; active involvement in recruitment across multiple teams; and contributing to the development of CoSyne's personal development framework. Additionally, hosted the nf-core community hackathon and spoke at Seqera Sessions, raising the team's external profile.

2025  
|  
2022

### Honorary Senior Research Fellow

University College London

📍 London, UK

- Acted as secondary supervisor to a PhD student and served on the thesis committee for another. Additionally contributed to analyses and provided input on papers across multiple research projects.

## CONTACT

✉ [rhreynolds@hotmail.co.uk](mailto:rhreynolds@hotmail.co.uk)

🐙 [GitHub](#)

in [LinkedIn](#)

📄 [ResearchGate](#)

## LANGUAGES

📊 R

🔗 Bash

🐍 Python

## TOOLS

🔖 Git/GitHub/GitLab

🔗 Nextflow

🐳 docker

*The source code is available  
[GitHub](#).*

*Last updated on 2026-08-17.*

2022  
|  
2021

## Research Fellow

University College London

📍 London, UK

- Lead analyst involved in processing and analysing transcriptomic data to identify molecular signatures of Parkinson's disease progression. Worked within a large, multi-PI team and coordinated with researchers to align analytical approaches. Work performed using R, Nextflow, and Docker.
- Co-lead of Code and Pipeline Alignment Working Group in the Aligning Sciences Across Parkinson's<sup>1</sup> initiative, facilitating the standardisation of data processing pipelines and coding practices across 10 international research teams, resulting in actionable recommendations to enhance data harmonisation and enable meta-analysis of post-mortem brain tissue datasets.
- Published 1 co-first author research article.

2016  
|  
2016

## Research Assistant

University of Copenhagen

📍 Copenhagen, Denmark

- Led project exploring the interactions between miR-34a, Sirt1 and p53 in a Huntington's disease mouse model, serving as lead experimentalist and analyst; work culminated in a first author publication<sup>2</sup>.
- In addition to these roles, facilitated collaboration with another research group and coordinated experiments with lab technicians, students, and researchers to support project goals.



## EDUCATION

2021  
|  
2016

### PhD, Bioinformatics

University College London

📍 London, UK

- Thesis: Exploring the importance of cell-type-specific gene expression regulation and splicing in Parkinson's disease<sup>3</sup>.
- Integrated GWAS summary statistics with bulk/single-cell transcriptomic, eQTL, and chromatin accessibility data to identify cell-type-specific regulatory mechanisms in Parkinson's disease. Methods used included partitioned heritability, COLOC, and MAGMA.
- Applied transcriptomic methods, including cell-type deconvolution, differential splicing analysis, and RNA-binding protein motif analysis, to investigate splicing alterations and their relevance to Lewy body diseases.
- Published 3 first/co-first author research articles and 1 first author review. Successfully secured £10,000 from Signe og Peter Gregersens Mindefond to undertake transcriptional profiling of Parkinson's disease brain tissue.

2016  
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2014

### MSc, Molecular Biomedicine

University of Copenhagen

📍 Copenhagen, Denmark

- Thesis: Changes in the miR-34a-SIRT1 axis in Huntington's disease
- Grade: A (92.5%)

2013  
|  
2010

### BSc, Molecular Biomedicine

University of Copenhagen

📍 Copenhagen, Denmark

- Thesis: Pro-apoptotic factors in Huntington's disease: a study in the R6/2 transgenic mouse model
- Grade: A (96.7%)



## TEACHING EXPERIENCE

2025  
|  
2021

### Subsidiary PhD Supervisor

University College London

📍 London, UK

- Secondary supervisor to Dr. Aaron Wagen. Provided support around transcriptomic analyses, bioinformatics workflows and statistical genetics.

2022  
|  
2019

### R fundamentals with Clinician Coders<sup>4</sup>

University College London

📍 London, UK

- Developed materials<sup>5</sup> and led workshops teaching basic R and tidy data principles to clinical academics.

2019  
|  
2017

### Omics Techniques

King's College London

📍 London, UK

- Lectured graduate level students on the principles of genome-wide association studies and led a workshop on how/why to use the Genotype-Tissue Expression portal.



## VOLUNTARY WORK

Present  
|  
2018

### Peer Reviewer

📍 London, UK

- Reviewer<sup>6</sup> for several scientific journals.

2022  
|  
2017

### Mentor

[Social Mobility Foundation](#)

📍 London, UK

- Mentored 4 A-level students looking to work in the field of biomedical research.



## KEY PUBLICATIONS

2023

### Local genetic correlations exist among neurodegenerative and neuropsychiatric diseases<sup>7</sup>

NPJ Parkinson's disease

- Reynolds, RH, Wagen, AZ, Lona-Durazo, F, Scholz, SW, Shoai, M, Hardy, J, Gagliano Taliun, SA, Ryten, M
- Role: Co-first author, lead analyst and corresponding author.
- Analysis<sup>8</sup> of local genetic correlations between neurodegenerative and neuropsychiatric disorders, with the aim of identifying genomic regions and genes that may drive pleiotropy.

2021

### Cross-platform transcriptional profiling identifies common and distinct molecular pathologies in Lewy body disorders<sup>9</sup>

Acta Neuropathologica

- Feleke, R, Reynolds, RH, Smith, A, Tilley, B, Gagliano Taliun, SA, Hardy, J, Matthews, PM, Gentleman, S, Owen, D, Johnson, MR, Srivastava, P, Ryten, M
- Role: Co-first author and analyst.
- Transcriptomic analysis<sup>10</sup> of cell-type-specific changes in the Lewy body diseases.

A full list of publications is available online at <https://rhreynolds.github.io/cv>

- 2019 **Informing disease modelling with brain-relevant functional genomic annotations<sup>11</sup>**  
Brain
- Reynolds, RH, Hardy, J, Ryten, M, Gagliano Taliun, SA
  - Role: First author.
  - Review of conceptual advances in the generation of brain-relevant functional genomic annotations and among tools that allow integration of these annotations with genome-wide association summary statistics.
- 2019 **Moving beyond neurons: the Role of cell type-specific gene regulation in Parkinson's disease heritability<sup>12</sup>**  
NPJ Parkinson's disease
- Reynolds, RH, Botía, JA, Nalls, MA, International Parkinson's Disease Genomic Consortium (IPDGC), System Genomics of Parkinson's Disease (SGPD), Hardy, J, Gagliano Taliun, SA, Ryten, M
  - Role: First author and lead analyst.
  - Analysis of Parkinson's disease common variation, with the aim of identifying cell types and pathways of importance to disease risk.

## CONFERENCES

- 2022 **AD/PD, Alzheimer's & Parkinson's Diseases Conference**  
 Hybrid event
- Talk: Identifying genetic correlations among neurodegenerative and neuropsychiatric diseases
- 2021 **Genomics of Brain Disorders**  
 Virtual event
- Talk: Dysregulation of splicing in human brain from individuals with Lewy body disease informs disease mechanisms
- 2019 **International Parkinson's Disease Genomics Consortium**  
 London, UK
- Talk: Pairing bulk and single-nuclear RNA-seq to identify dementia-related pathways in PD
- 2019 **AD/PD, Alzheimer's & Parkinson's Diseases Conference**  
 Lisbon, Portugal
- Talk: Mapping Parkinson's disease heritability to specific brain cell types
  - Received mention in a blog post on Alzforum<sup>13</sup>.
- 2018 **International Parkinson's Disease Genomics Consortium**  
 Reykjavik, Iceland
- Talk: Moving beyond neurons: exploring the importance of cell type-specific gene expression in Parkinson's disease

## LINKS

- 1: <https://parkinsonsroadmap.org/research-network/pd-functional-genomics/>
- 2: <https://pubmed.ncbi.nlm.nih.gov/29289683/>
- 3: <https://discovery.ucl.ac.uk/id/eprint/10119171/>

- 4: <https://www.ucl.ac.uk/school-life-medical-sciences/about-slms/office-vice-provost-health/academic-careers-office/career-schemes/clinician-coders>
- 5: <https://github.com/ClinicianCoders/ClinicianCoders>
- 6: <https://publons.com/researcher/3017104/regina-hertfelder-reynolds/peer-review/>
- 7: <https://pubmed.ncbi.nlm.nih.gov/37117178/>
- 8: <https://rhreynolds.github.io/neurodegen-psych-local-corr/>
- 9: <https://pubmed.ncbi.nlm.nih.gov/34309761/>
- 10: <https://rhreynolds.github.io/LBD-seq-bulk-analyses/>
- 11: <https://pubmed.ncbi.nlm.nih.gov/31603214/>
- 12: <https://pubmed.ncbi.nlm.nih.gov/31016231/>
- 13: <https://www.alzforum.org/news/conference-coverage/expression-expression-expression-time-get-board-eqtls>